

6.2 Deformation: Elastic & Plastic Behaviour

Question Paper

Course	CIEA Level Physics
Section	6. Deformation of Solids
Topic	6.2 Deformation: Elastic & Plastic Behaviour
Difficulty	Easy

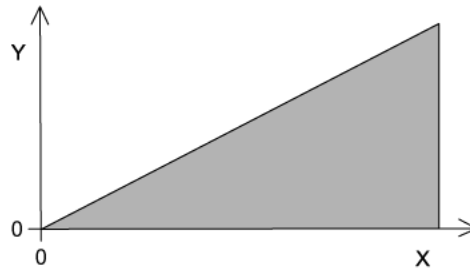
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student experimented on a metal wire; the graph below was plotted.



The area under the graph shows the total strain energy stored in the wire when stretched.

How should the axes be labelled?

	Y	X
A	mass	extension
B	force	extension
C	strain	energy
D	stress	strain

[1 mark]

Question 2

Four rods were made from four different materials; all the rods had the same dimensions.

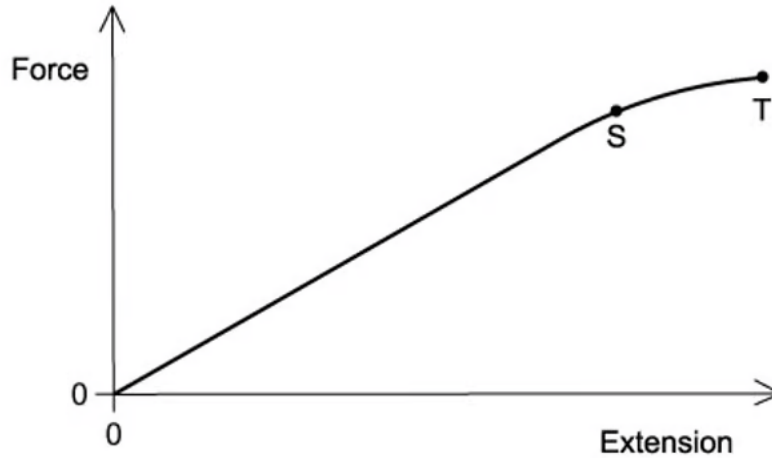
At room temperature, which of the following would sustain the biggest plastic deformation?

- A. the ductile material steel
- B. the brittle material glass
- C. the brittle material carbon
- D. the ductile material aluminium

[1 mark]

Question 3

A student drew a force-extension graph for a sample of metal. She subjected the sample to a force that increased to a maximum then decreased to zero.



When the sample contracts it follows the same force-extension curve as when it was being stretched.

What is the behaviour of the metal between S and T?

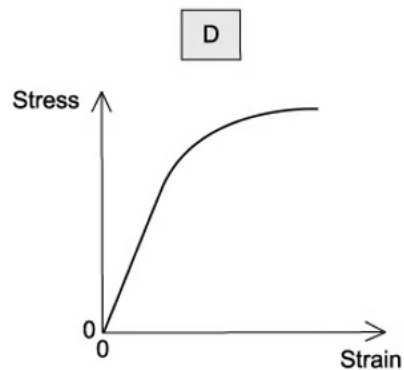
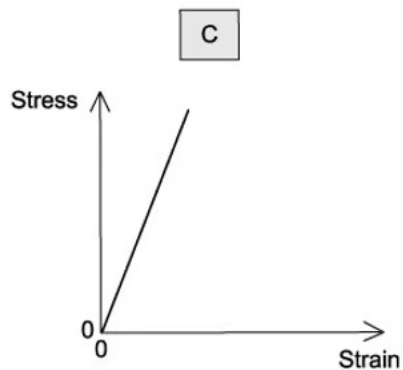
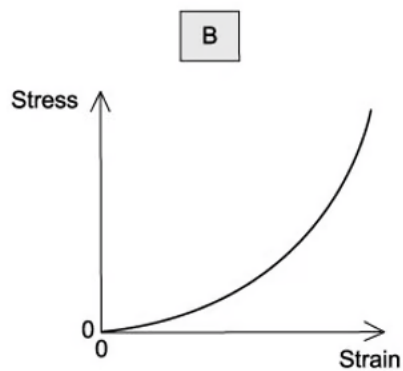
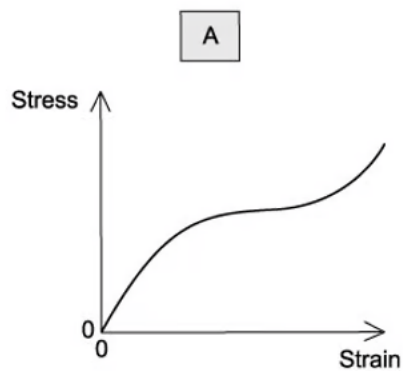
- A. both elastic and plastic
- B. elastic but not plastic
- C. not elastic and not plastic
- D. plastic but not elastic

[1 mark]

Question 4

Graphs of the stress-strain for four different materials are shown below.

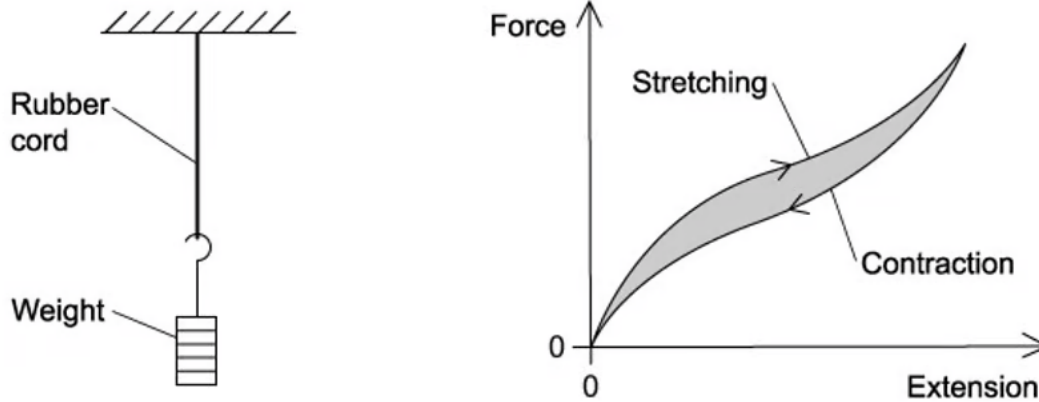
Which one shows the correct graph for a ductile metal?



[1 mark]

Question 5

A weight was attached to the lower end of a length of rubber. The rubber was hung from a solid support. The weight was gradually increased from zero then gradually reduced to zero.



The graph shows the force-extension curve for contraction (below) and the force-extension curve for stretching.

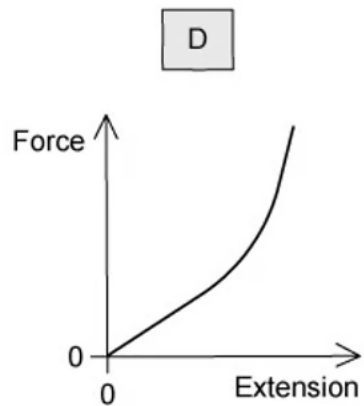
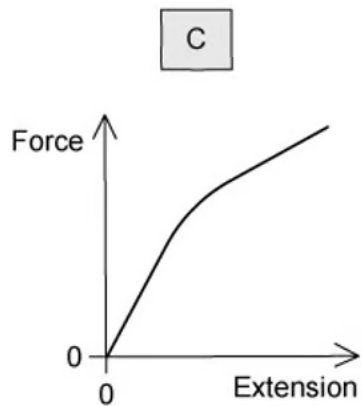
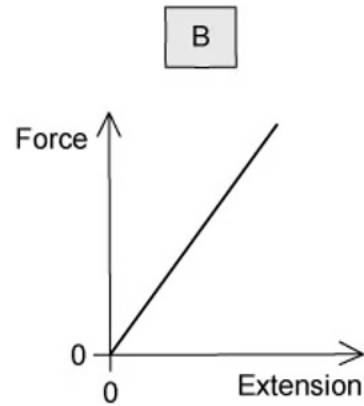
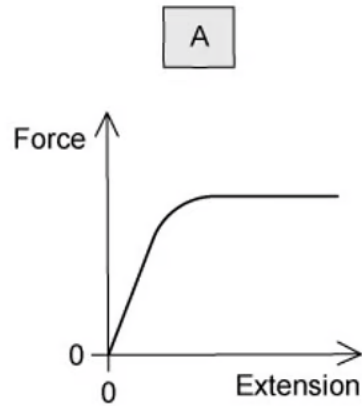
What does the shaded area between the curves represent?

- A. the amount of elastic energy stored in the rubber
- B. the work done by the rubber cord during contraction
- C. the work done on the rubber cord during stretching
- D. the amount of thermal energy dissipated in the rubber

[1 mark]

Question 6

The graphs below are force-extension graphs, which one is a rubber band this is stretched almost to its breaking point?



[1 mark]

Question 7

A metal wire had a tension F and extension x with a Young modulus of E ; the wire obeyed Hooke's law

Which expression represents the elastic strain energy stored in the wire?

A. $\frac{1}{2}Fx$

B. Ex

C. $\frac{1}{2}Ex$

D. Fx

[1 mark]

Question 8

Which of the following is the correct definition for elastic and plastic behaviour?

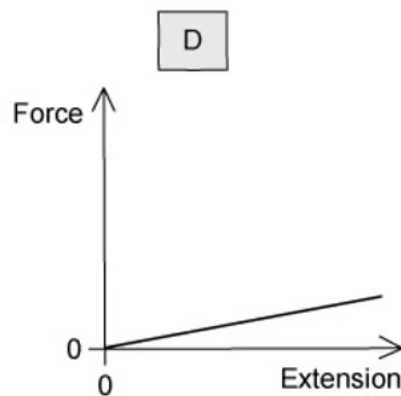
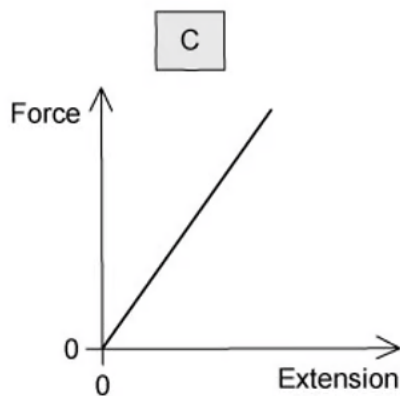
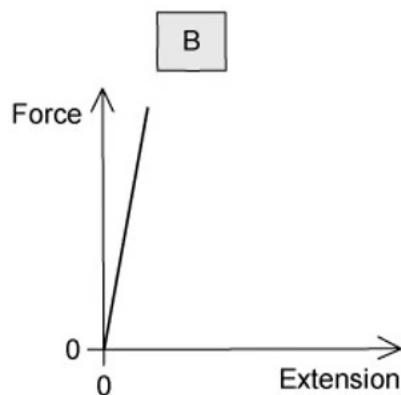
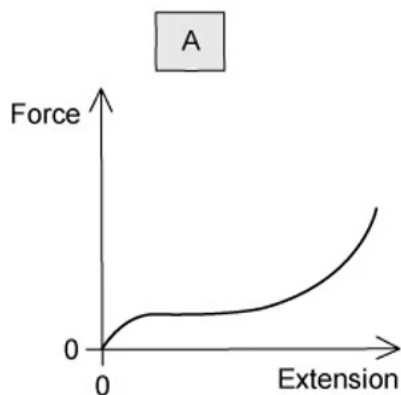
	Elastic behaviour of a material	Plastic behaviour of a material
A	returns to its original shape and size	suffers permanent deformation
B	obeys Hooke's Law	extends continuously under a steady load
C	has a linear force-extension curve	has a horizontal force-extension curve
D	extends only within the limit of proportionality	extends beyond the limit of proportionality

[1 mark]

Question 9

The graphs below all show force-extension graphs; they are all drawn to the same scale.

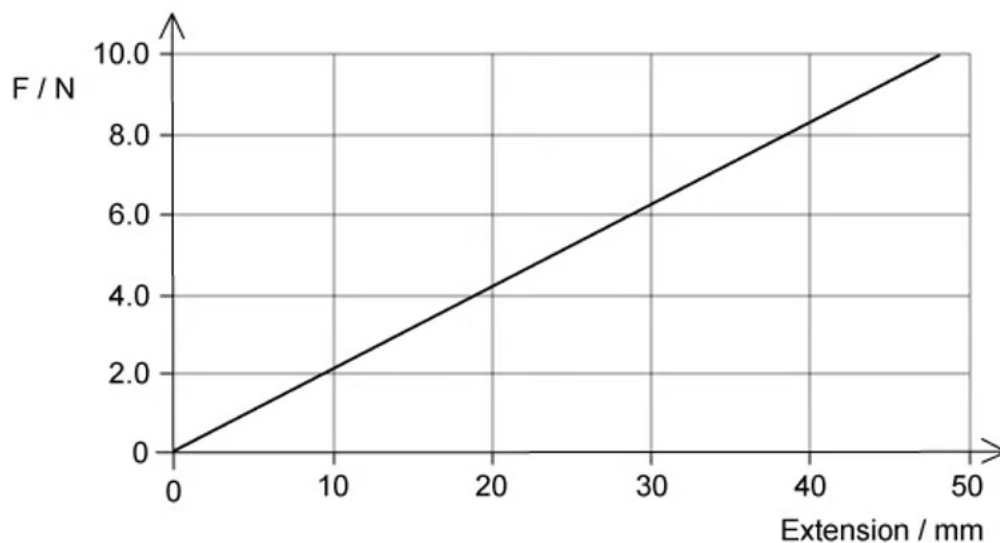
Which graph shows the object that has been deformed and has the greatest amount of elastic potential energy?



[1 mark]

Question 10

A spring is extended with varying forces; the graph below shows the results.



What is the energy stored in the spring when the extension is 30 mm?

- A. 0.25 J
- B. 0.95 J
- C. 0.095 J
- D. 0.19 J

[1 mark]

CIE A Level Physics

Topic Questions



Model Answers: Easy

1

The correct answer is **B** because:

- Strain is the measurement of how much an object is stretched
- Strain energy is equal to $\frac{1}{2}kx^2 = \frac{1}{2}Fx$
- The question states that the area under the graph shows the total strain energy stored
- For this to be true, the axis must be:
 - X = extension x
 - Y = force F

2

The correct answer is **D** because:

- Brittle materials break easily under tension
- Ductile materials are solids that can deform under tensile stress
- Aluminium can sustain a larger plastic deformation than steel, so the correct answer is **D**

A is incorrect because steel is ductile, but aluminium can sustain a larger plastic deformation

B & C are incorrect because glass and carbon are examples of brittle materials and will break easily with a small tension

3

The correct answer is **B** because:

- Force-extension graphs tell us about the strain on a material
- Elasticity is the ability of the material to resist a permanent change, that is it will return to its original shape and size after the stress is removed
- Plasticity is a non-reversible change of shape in response to a force
 - Point S on the graph is the limit of proportionality
 - Point T is the elastic limit
- After the elastic limit, the material would exhibit plastic behaviour, meaning it would have a permanently changed shape
- Therefore, as the sample has not gone past point T, the material would be exhibiting elastic behaviour, but not plastic behaviour

4

The correct answer is **D** because:

- Brittle materials break easily under tension
- Ductile materials are solids that can deform under tensile stress
- A ductile material will withstand large plastic strain, whereas a brittle material will not
- A metal wire tends to be described as ductile as it will experience a large amount of extension when subjected to stress, especially beyond the elastic limit



- A ductile metal will obey Hooke's law initially, which is represented as a straight line on the graph
- The metal would then exhibit plastic behaviour after the limit of proportionality, which is represented on the graph as a curve

A is incorrect because the graph shows an unusual shape stress-strain graph associated with a polymer, such as rubber

B is incorrect because the graph doesn't show proportionality

C is incorrect because the graph shows a brittle material that shows no plastic strain

5

The correct answer is **D** because:

- The area under the force-extension graph shows the strain energy (work done)
 - Strain energy is equal to $= \frac{1}{2}Fx$
- The area under the stretching graph is the energy used in stretching the rubber
- The area under the contraction graph is less than the area of the stretching graph
- The difference is shown in the shaded area between the two
 - This is the thermal energy which is dissipated to the surroundings

A is incorrect because the amount of elastic energy stored in the rubber is the total area under both curves

B is incorrect because the work done by the rubber cord during contraction is represented by the area under the lower curve

C is incorrect because the work done on the rubber cord during stretching is represented by the area under the upper curve

6

The correct answer is **D** because:

- The force-extension graph shows the strain energy (work done)
 - Strain energy is equal to $\frac{1}{2}Fx$

- Rubber is a polymer and when fully stretched the long-chain molecules are fully lined up so the rubber will not stretch any more
 - Rubber deforms elastically up to a certain point as the polymers chains, which were previously curled up, uncurl
- Graph **D** shows that once the band is stretched fully, it takes lots of force to increase the extension

A is incorrect because the graph suggests that once stretched no extra force is needed to extend the rubber

B is incorrect because the graph shows a material that obeys Hooke's law

C is incorrect because the graph suggests that less force is needed to extend the rubber band even for a small extension

7

The correct answer is **A** because:

- The question states that Hooke's law is obeyed so that graph would be a straight line passing through the origin
- The triangular area under the graph represents the elastic strain energy
- Therefore, the equation for elastic strain energy is $= \frac{1}{2}Fx$

8

The correct answer is **A** because:

- Elasticity is the ability of the material to resist a permanent change, that is it will return to its original shape and size after the stress is removed
 - Hence, an elastic material will return to its original shape and size
- Plasticity is a non-reversible change of shape in response to a force
 - Hence, a plastic material will suffer a permanent deformation

B is incorrect because materials with plastic behaviour will not extend continuously

C is incorrect because not all elastic materials will show a linear force-extension curve, consider rubber

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 - Hence, a plastic material will suffer a permanent deformation

B is incorrect because materials with plastic behaviour will not extend continuously

C is incorrect because not all elastic materials will show a linear force-extension curve, consider rubber

D is incorrect because the definition of a material obeying Hooke's law is that extends only within the limit of proportionality

9

The correct answer is **C** because:

- Elastic potential energy is equal to $\frac{1}{2}Fx$
- It is represented by the area under the force-extension graph
- The graph that shows the largest area under the line or curve is **C**

10

The correct answer is **C** because:

- Elastic potential energy $\frac{1}{2}Fx$
- From the graph
 - When the extension is $x = 0.03 \text{ m}$
 - Force is equal to $F = 6.3 \text{ N}$
 - Strain energy $= \frac{1}{2} \times 6.3 \times 0.03 = 0.0945 \text{ J}$
- Therefore, the energy stored in the spring is 0.095 J